Non Verbal Reasoning / Symbol Series / Pattern Recognition

Types of Questions

1. Series
2. Classification
3. Mirror Images
4. Embedded Images
5. Figure Matrix
6. Paper Cutting
7. Grouping of Images
8. Shape Construction
9. Cubes and Dices
10. Analogy
11. Water Images
12. Pattern Completion
13. Paper Folding
14. Rule Detection
15. Dot Situation
16. Image Analysis

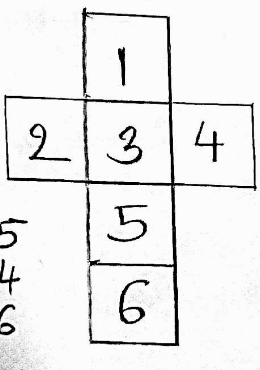
The perfect trick to solve Non Verbal Reasoning Questions is to Eliminate Options.

Cubes and Dices

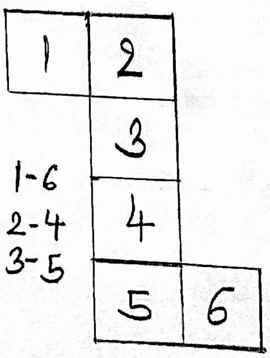
Opposing sides

~~1 – 5~~ 1 – 6
~~2 – 4~~ 2 – 5
~~3 – 6~~ 3 – 4

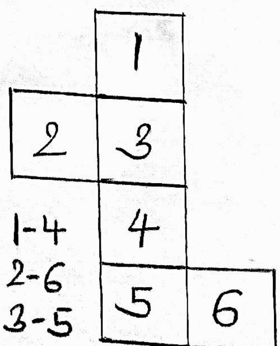
Patterns : (The numbers given in patterns are not "real" numbers of dice, they are just representing blocks)



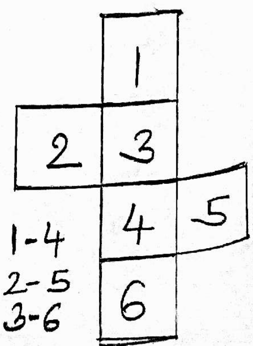
1-5
2-4
3-6



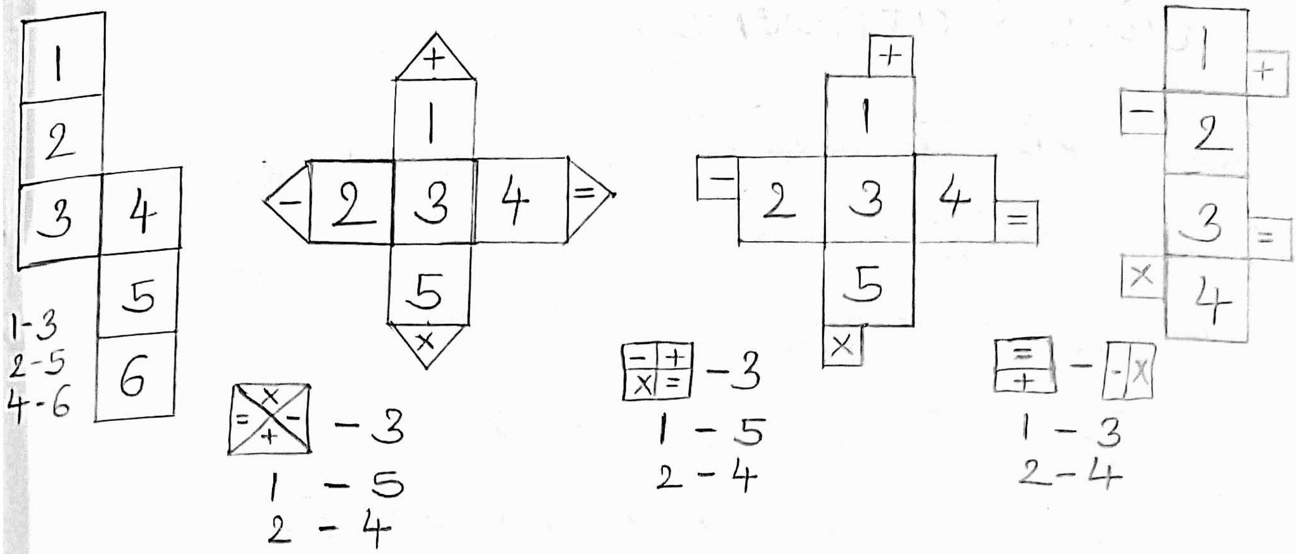
1-6
2-4
3-5



1-4
2-6
3-5



1-4
2-5
3-6



Number Letter Series

Types of Patterns in

Number Series

- Difference
- Double Difference
- Square
- Cube
- Prime Numbers
- Multiples
- Factors
- Series with combinations
- Miscellanous

Alpha-Numeric

A	a	1
B	b	2
C	c	3
D	d	4
E	e	5
F	f	6
G	g	7
H	h	8
I	i	9
J	j	10
K	k	11
L	l	12
M	m	13
N	n	14
O	o	15
P	p	16
Q	q	17
R	r	18
S	s	19
T	t	20
U	u	21
V	v	22
W	w	23
X	x	24
Y	y	25
Z	z	26

Conjugate Alphabets

A B C D E
Z Y X W V

F G H I J
U T S R Q

K L M N O
P O N M L

P Q R S T
K J I H G

U V W X Y
F E D C B

Z
A

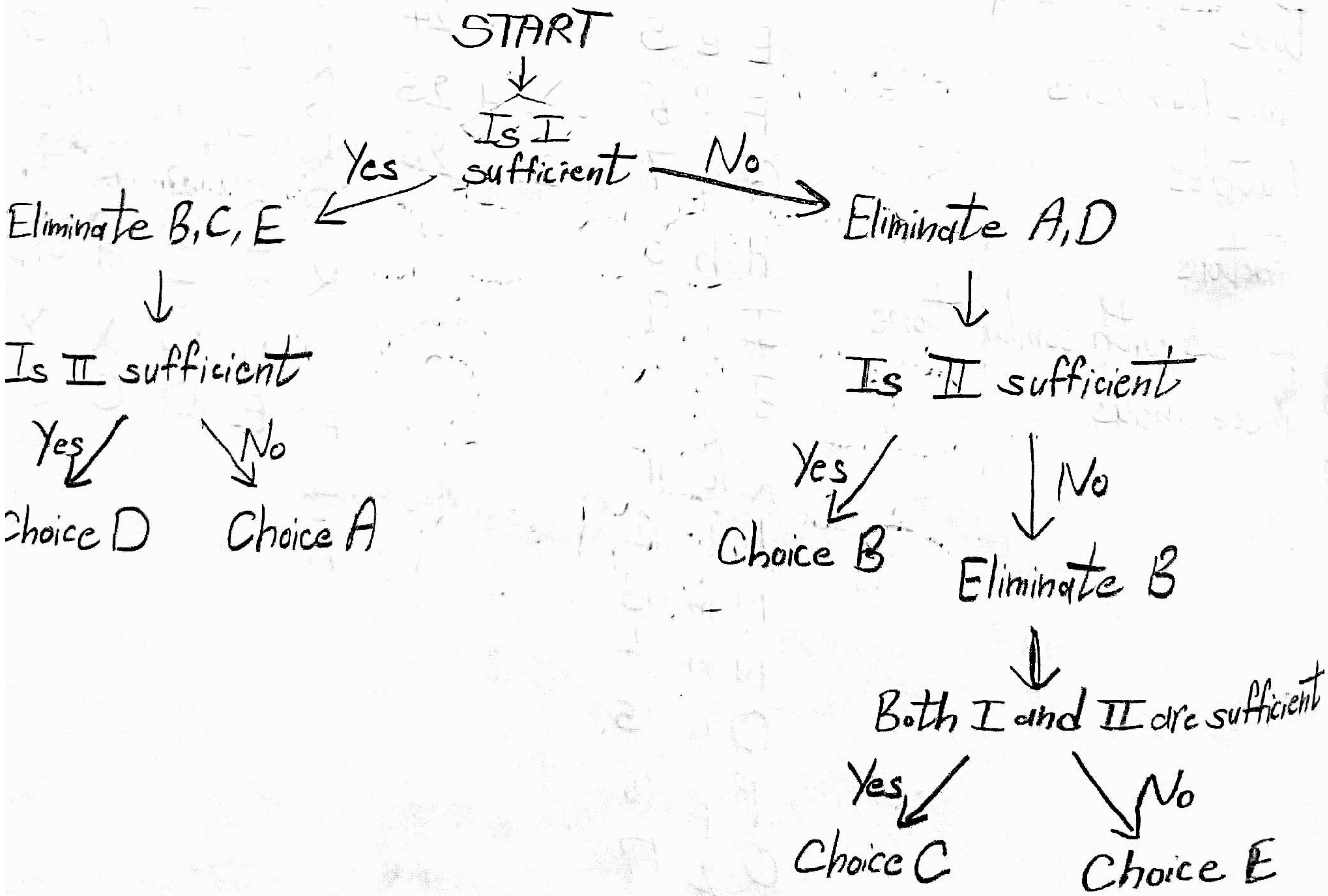
Data Sufficiency

Statement I :

Statement II :

a) Statement I alone is sufficient.
b) Statement II alone is sufficient.
c) Both statements put together are sufficient.
d) Either of the statements individually is sufficient.
e) Both statements put are not sufficient.

Tips :
- We are looking for unique solution
- Search the question type [(i) YES/NO (ii) Value Based]
- When looking at statement 2, forget statement 1 and vice versa.
- NO is as good as YES.
- Answer in the first sight is wrong most of the times, so take time and recheck.

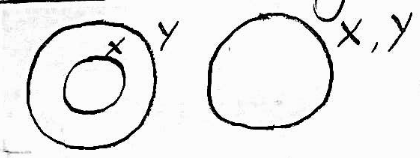
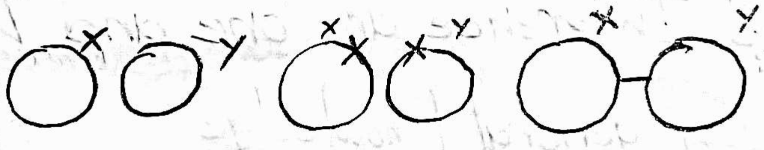
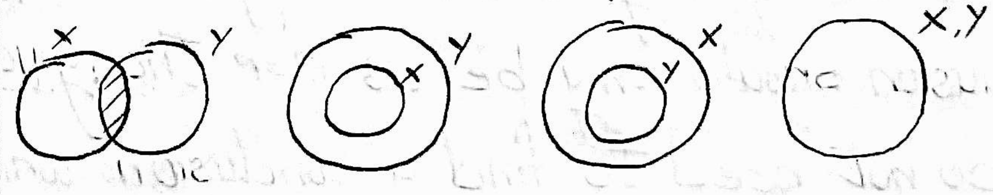
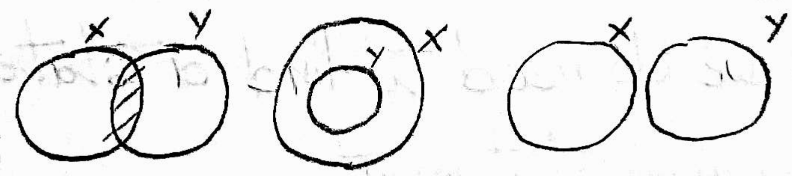


Syllogism

The question contains some statements followed by some conclusions. We need to decide which of the given conclusions logically follow from the given statements, disregarding commonly known facts.

Steps :

1. Draw the basic diagram first with minimum overlap.
2. For a statement to be false, if it is false in even any one diagram, then also it is false.
3. For a statement to be true, it must be true in all possible diagrams.
4. For a statement to be possibility, it must be true in even any one diagram.

Statement	Venn Diagram
All X's are Y's	
No X's are Y's	 All three are same just different ways of writing
Some X's are Y's	
Some X's are not Y's	

Logical Deduction and Deductive Reasoning

In a statement-conclusion, a paragraph or a statement is given to us and based on that we must derive certain conclusions.

Now what is a conclusion? So, a conclusion is a judgement or a decision which is derived after sufficient reasoning. In our context, conclusions are logical facts that are derived or inferred from the passage.

Steps To Solve :

- Read the passage or given statements thoroughly.
- Analyze each statement individually.
- Draw a logical conclusion based on the given statement.

Some Important Points

- Conclusion and Inference are one and the same thing.
- Do not use your general knowledge.
- Conclusion should only be as per the given statement, so we do not need to find a conclusion which is true in general, but we do need to find a statement which is true for the given statement.

Tip No. 1 : Do Not Assume

Only make those assumptions that can be drawn or inferred from the given passage or statements. Do not assume anything which is not related to the passage.

Tip No. 2 : Advice or Result

If a given conclusion is in advice form or a certain result can be deduced out of it then such a conclusion will always be true. But of course, the conclusions should be related to the given statement.

Tip No. 3 : Avoid pre-assumptions

If a given conclusion contains a "Pre-Assumption" then such a conclusion is always wrong.

Tip No. 4 : Do Not Go By Morals / Ethics

For certain questions, when you go by the options for finding out an appropriate conclusion, sometimes it may happen that the answer option may not be morally correct, still that will be the correct answer. So, if an option is related to the statement and if you find it ~~bo be~~ to be it logically correct then that option should be selected even it is morally incorrect.

Statement and Argument

Strong Argument	Weak Argument
Security of nation	Over exaggeration, Assumption
Development of a country	Words like Each, Nothing, Every, All, Only, None, No, one
Universal Truth	Comparison
Instructions from Supreme Authority	Ambiguous, Unclear
Facts based on experience and proof	Half explained arguments
Educational Development	
Advice,	Morals, Ethics, General Knowledge ; Not Practical, Irrelevant to statement
If Advantage and Disadvantage leads in same direction.	If Advantage and Disadvantage leads in opposite direction, i.e., we have to choose between Advantage and Disadvantage ("OR" word)

Statement and Assumption

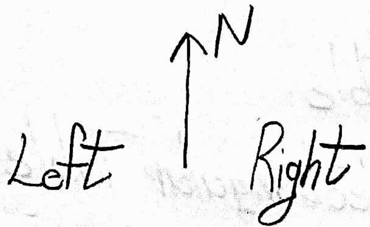
Implicit means true
Explicit means false

1. Words like EVERY, EACH, NONE, ALL, ONLY, etc., will always lead to false assumption.
2. Any assumption which is not sticked to statement is FALSE
3. Any assumption which talks about past or future is FALSE
4. Any assumption which goes against the given statement is FALSE.
5. If statement is generalised and assumption is specific or vice-versa, then assumption is FALSE.

Seating Arrangement

In this topic, there is lot of raw data given. Based on this data certain conclusions are to be drawn. The given raw data will help us to arrange the objects or people in a predetermined manner after performing necessary analysis over it.

If a person is facing North, then "Your left and right" will be the same as the "Person facing North".



If a person is facing South, then it will be a mirror image, i.e., "Person's Left" will be "Your Right" and vice-versa.

Right ↓ Left

In "Circular Seating Arrangement" when all the persons are facing inwards or ~~outwards~~ center –

Persons to his/her (specific person) right ~~is~~ are in anti-clockwise direction.

Persons to his/her (specific person) left are in clockwise direction.

Whereas when all the persons are facing outwards then

Persons to his/her (specific person) left are in anti-clockwise direction.

Persons to his/her (specific person) right are in clockwise direction

Tricks to solve questions :

Step 1 : First go through the entire question and count total number of people and prepare a "Seating Frame".

Step 2 : Solve individual statements and then merge all the solved small sets into a final complete set.

In seating arrangement, questions are mainly based on 4 main types :

– Single Row — Circular Table

– Double Row — Square or Rectangular Table

Word "Who" is used for person who is in left of the word "who". Words "And" and "Is" are used for first person given in the statement.

Blood Relation

———————— Great grandfather, Great grandmother

———————— Grandfather, Grandmother

———————— Father, Mother, Uncle, Aunt

your generation → ———————— You, Brother, Sister, Wife, Husband

———————— Kids, Son, Daughter, Nephew, Niece

———————— Grandson, Granddaughter

Types of Relation :

1. Direct and Indirect :

Direct : Direct Blood Relationships and Marriage.
For ex, Father-Daughter, Father-Son, Husband-Wife

Indirect : Not directly related by blood and more than one generation.
For ex, Nephew, Niece, Uncle, Aunt, Grandfather - Grandson

② Transparent and Opaque

Transparent : All relations except with cousins.

Opaque : Relations between Cousins

③ Paternal and Maternal

Paternal : Father's side

Maternal : Mother's side

Drawing Relationship Chart

① Write + on male and - on female to identify gender.

② Write people in same generation in same line

③ Use — between spouses, | while crossing generation and = between siblings.

④ — : Husband - Wife

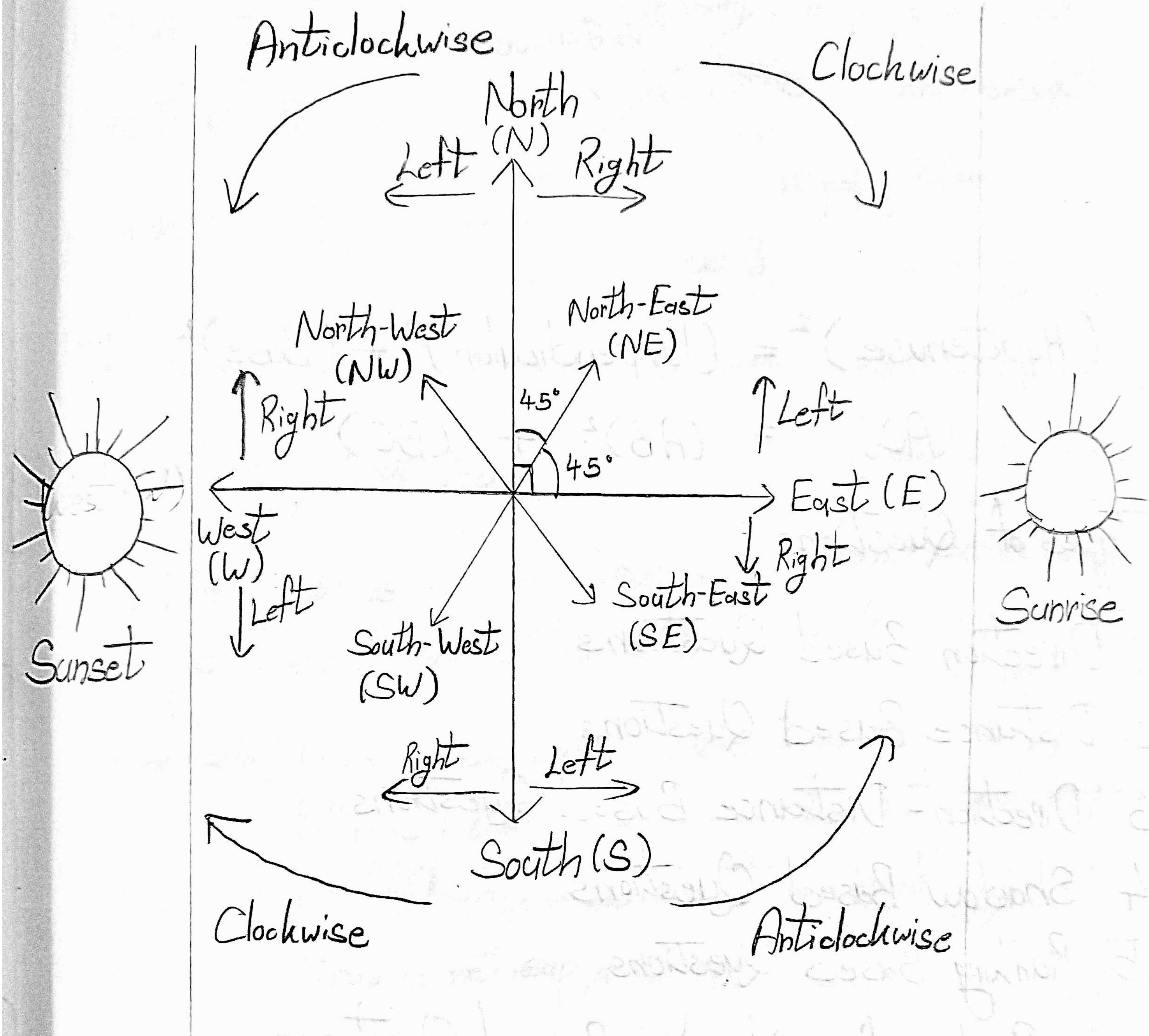
| : Father-Son, Father - Daughter, ~~Grath~~ Grand father - Father

= : Brother - Sister, Brother - Brother, Sister - Sister

Types of Questions

- General Questions
- Coded Blood Relation
- Indicating Blood Relation
- Puzzle Type Questions

Distance Direction Ranking



Main Directions : North, South, East, West

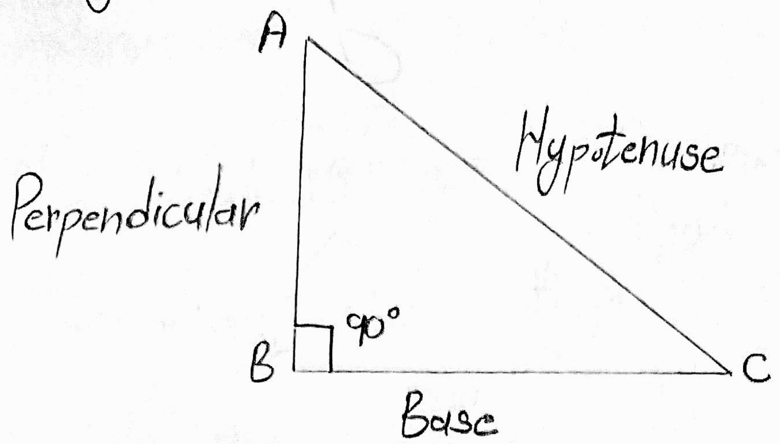
Cardinal Directions : North-East, North-West, South-East, South-West

Clockwise and Anticlockwise directions never changes

Selection of right and left direction depends on whether it is clockwise (Right turn) or anti-clockwise (Left turn).

Right ⟶ Clockwise, Left ⟶ Anticlockwise

Pythagoras Theoram:



$$(\text{Hypotenuse})^2 = (\text{Perpendicular})^2 + (\text{Base})^2$$

$$(AC)^2 = (AB)^2 + (BC)^2$$

Types of Questions:

1. Direction Based Questions
2. Distance Based Questions
3. Direction - Distance Based Questions
4. Shadow Based Questions
5. Ranking Based Questions.

Key Points for Shadow Based Questions

1) There would be no shadow at 12 Noon.
2) After sun rise till 12 PM, shadow would be towards west.
3) After 12 PM and before sunset, shadow would be towards east
4) If a person faces towards North before 12 pm, his/her shadow would be towards his/her left side and after 12pm it would be towards his/her right side.
5) If a person faces towards South before 12pm, his/her shadow would be towards his/her right side and after 12 pm it would be towards his/her left side.
6) Right and Left turn always make 90° angle.

Calendar

Year — 1
Manths — 12
Weeks — 52
Days — 365/366

	Ordinary Year /	Leap Year
	365 days	366 days
	$7 \times 52 = 364$	
	1 odd day	2 odd days

Identifying Leap Year

If the year is century, leap year should be divisible by by 400. Else it should be divisible by 4.

For example, 1996 is leap year because it is divisible by 4
2019 is not leap year because it is not divisible by 4.
1600 is leap year because it is divisible by 400.
1700 is not leap year because it is not divible by 400.

Day	Code
Sunday	0
Monday	1
Tuesday	2
Wednesday	3
Thursday	4
Friday	5
Saturday	6

Q. If today is Monday, then what will be the day on 23^{rd} day?

$\Rightarrow \frac{23}{7} \Rightarrow$ Remainder $= 2$

Day for Code 2 is Tuesday, so answer is Tuesday

If suppose in question, it gives that today is Thurday, then we consider ~~tod~~ Thursday as 1 ~~as~~ and remaining codes will be written in consecutive order.

Concept of Odd Days

In Year 100,

100 Years

$\frac{100}{4} = \cancel{25}$ 24 Leap Years

100 is not a leap year as it is not divisible by 400.

76 ordinary years $\xrightarrow{76 \times 1}$ 76 odd days

24 leap years $\xrightarrow{24 \times 2}$ 48 odd days

Removing by dividing by 7 (a week) and taking remainder

$76 \% 7 = 6$

$48 \% 7 = 6$

6 odd days + 6 odd days

↓

12 odd days

$12 \% 7 = 5$

→ 5 odd days

Year	Number of odd days
100	5
200	3
300	1
400	0
All multiples of 400	0

By this table, we can easily find number of odd days of any year

Odd days in Year,

$500 = 400 + 100 = 0 + 5 = 5$ odd days

$1800 = 1600 + 200 = 0 + 3 = 3$ odd days

Q. On what day does 1 January 1971 lie ?

⇒ First find odd days in 1970,

$1970 = 1600 + 300 + 70$

$= 0 + 1 + 3$

$= 4$ odd days

Calculation here

$\frac{70}{4} = 17$

70 Years

53 ordinary years → 53×1 → 53 odd days → $53 \% 7$ → 4 odd days

17 leap years → 17×2 → 34 odd days → $34 \% 7$ → 6 odd days

4 odd days + 6 odd days

↓

10 odd days

↓ $10 \% 7 = 3$

3 odd days

This is till 1970. Considering 1st of January 1971, we will have 5 odd days.

Code for Friday is 5.

So answer is 5

Q. On what day does 22 October 1964 lie?

⇒ First we will find odd days in 1963.

$$1963 = 1600 + 300 + 63$$
$$= 0 + 1 + 1$$

Year 1963 = 2 odd days

$\frac{63}{4} = 15$ leap years

Odd Days in Months with
31 days → 3
30 days → 2
29 days → 1
28 days → 0

63 year → Same as year → 63 odd days → $63 \% 7$ → 0 odd days

63 year → Number of leap years → 15 odd days → $15 \% 7$ → 1 odd day

0 odd days + 1 odd day → 1 odd day

1963 → 2 odd days

Jan 1964 → 3 odd days

Feb 1964 → 1 odd day (leap year)

March 1964 → 3 odd days

April 1964 → 2 odd days

May 1964 → 3 odd days

June 1964 → 2 odd days

July 1964 → 3 odd days

August 1964 → 3 odd days

September 1964 → 2 odd days

October 1964 → $22 \% 7 = 1$ odd day

∴ Adding all odd days in 1963 and till 22 October 1964, we have 25 odd days.

$25 \% 7 =$ ~~2 odd days~~ 4 odd days

Code for Thursday is 4, hence, answer is 4.

Months Starting with Same Day

Ordinary Year	Leap Year
January (31) – October (31)	January (31) – April (30) – July (31)
February (28) – March (31) – November (30)	February (29) – August (31)
April (30) – July (31)	March (31) – November (30)
September (30) – December (31)	September (30) – December (31)

Q. For a ordinary year (non-leap year), how many month-pairs in a year have 31 days and start and end on the same day?

⇒ January - October

Q. For a leap year, how many month-pairs in a year have 31 days and start and end on the same day?

⇒ January - July

Q. For a ordinary year (non-leap year), how many months-pair in a year have 30 days and start and end on the same day?

⇒ January - October

Though January and October have 31 days in total, they also have 30 days. Problem with other pairs is that they don't end at same day.

Finding Day on a particular Date when Day on some other Date is given to us

It has Four Categories

a) Month and Year Same, Date Different

1st March 2011 ⟶ Monday (Future 1 ⟶ 31)

31st March 2011 ⟶ ?

$$\begin{array}{r} 31 \\ -1 \\ \hline 30 \end{array}$$

30 % 7 = 2 odd days

31st March 2011 ⟶ Monday + 2 = Wednesday

31^{st} March 2011 → Wednesday (Past 31 → 1)
1^{st} March 2011 → ?

$$\begin{array}{r} 31 \\ -1 \\ \hline 30 \end{array} \quad 30\%7 = 2 \text{ odd days}$$

1^{st} March 2011 → Wednesday − 2 = Monday

b) Date and Month Same, Year Different

5^{th} January 2001 → Monday (Future 2001 → 2007)
5^{th} January 2007 → ?

$$\begin{array}{r} 2007 \\ -2001 \\ \hline 6 \text{ years} \end{array} \quad \frac{6}{4} = 1 \text{ leap year}$$

6 years → 5 ordinary years + 1 leap year

5 ordinary years $\xrightarrow{5\times1}$ 5 odd days

1 leap year $\xrightarrow{1\times2}$ 2 odd days

5 odd days + 2 odd days → 7 odd days $\xrightarrow{7\%7}$ 0 odd days

5^{th} January 2007 → Monday + 0 = Monday

15^{th} March 1848 → Monday
15^{th} March 1862 → ?

In this question, we have 1862 − 1848 = 16 years difference and 14/4 = 3 leap years. But 1848 will not be considered as leap year as data is given after February. So leap years are 1852, 1856 and 1860.

15^{th} January 1848 $\longrightarrow$ Monday

15^{th} January 1862 $\longrightarrow$?

$$\begin{array}{r} 1862 \\ -1848 \\ \hline 14 \end{array}$$ years difference

$\frac{14}{4} = 3$ leap years ✗

4 leap years as 1848 will be considered

14 years

~~11 ordinary years~~ ~~3 leap years~~

10 ordinary years $\xrightarrow{10 \times 1}$ 10 odd days $\xrightarrow{10 \% 7}$ 3 odd days

4 leap years $\xrightarrow{4 \times 2}$ 8 odd days $\xrightarrow{8 \% 7}$ 1 odd day

3 odd days + 1 odd day $\rightarrow$ 4 odd days

15 January 1862 $\longrightarrow$ Monday + 4 = Friday

c) Date and Year Same, Month Different

23^{rd} March 2015 $\longrightarrow$ Tuesday

23^{rd} November 2015 $\longrightarrow$?

Go ~~t~~ till one month before final month

March $\longrightarrow$ 3 odd days

April $\longrightarrow$ 2 odd days

May $\longrightarrow$ 3 odd days

June $\rightarrow$ 2 odd days

July $\longrightarrow$ 3 odd days

August $\longrightarrow$ 3 odd days

September $\longrightarrow$ 2 odd days

October $\longrightarrow$ 3 odd days

Total odd days = 21 $\quad 21 \% 7 = 0$ odd days

23^{rd} November 2015 $\longrightarrow$ Tuesday + 0 = Tuesday

<u>Another Method</u>:

2015 is an ordinary (non-leap) year. In this year, March and November start on same day. Hence their 23^{rd} date will be also on same day.

16th January 1992 ⟶ Monday
16th December 1992 ⟶ ?

1992 is a leap year. We know that in a leap year, January and July start and end at same day. Also September and December start on same day. So 16 September 1992 and 16 December 1992 will also be on same day.
So we have reduced question to,

16th July 1992 ⟶ Monday
16th September 1992 ⟶ ?

Go till one month before final month

July ⟶ 3 odd days
August ⟶ 3 odd days
Total 6 odd days

16th December 1992 ⟶ 16th September 1992 ⟶ Monday + 6
= Sunday

d) All three different

10th March 2013 ⟶ Monday
27th August 2017 ⟶ ?

First make Year same, then Date, then Month.
We will first calculate from 10th March 2013 to 10th March 2017

$$\begin{array}{r} 2017 \\ -\,2013 \\ \hline 4 \end{array}$$ years

3 ordinary years, 1 leap year

4

Total Odd Days = 5

10th March 2017 ⟶ Monday + 5 = Saturday

Now we will calculate from 10^{th} March 2017 to 27^{th} March 2017

$$17 \% 7 = 3 \text{ odd days}$$

$$\begin{array}{r} 27 \\ -10 \\ \hline 17 \end{array} \text{ days difference}$$

27^{th} March 2017 $\longrightarrow$ Saturday + 3 = Tuesday

Now we will calculate from 27^{th} March 2017 to 27^{th} August 2017

We will go ~~one~~ till one month before final month.

March $\longrightarrow$ 3 odd days
April $\longrightarrow$ 2 odd days
~~Ma~~ May $\longrightarrow$ 3 odd days
June $\longrightarrow$ 2 odd days
July $\longrightarrow$ 3 odd days

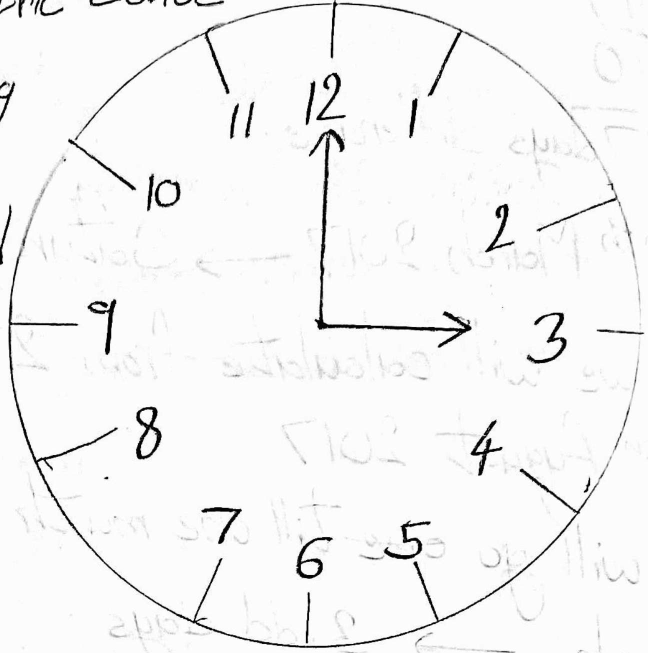
Total odd days = 13

$$13 \% 7 = 6 \text{ odd days}$$

27^{th} August 2017 $\longrightarrow$ Tuesday + 6 = Monday

Clocks

Clock is a mechanical or electric device for measuring time, indicating hours, minutes, and sometimes seconds by hands on a round dial or by displayed figures



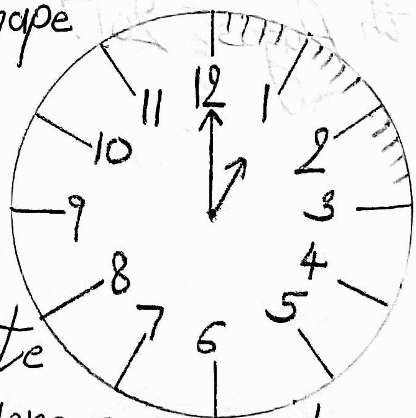
Hour Hand is also known as Short Hand because it is smaller hand in Clock.

Minute Hand is also known as Long Hand ~~bae~~ because it is larger hand in Clock.

→ Hour Hand (Short Hand)
→ Minute Hand (Long Hand)

Questions are asked ~~from~~ on mechanical type of clock (Not from digital clock/watch)

Key Points



A. The dial of the clock is circular in shape and is divided into 60 equal parts, called minute spaces.

B. 60 minute spaces trace an angle of 360°.

C. In 1 hour, Minute hand traverses 60 minute space or 360° degree, Hour hand traverses 5 minute space or 30° degree.

D. In 1 minute, Minute hand traverses 6° (dpm - Degree per Minute) Hour hand traverses 0.5° dpm (Degree per Minute).

E. Degree between each number on the clock is 30°.

Important types of questions:
1. Angle between hour and minute hand
2. Mirror Image
3. Water Image

Angle between hour and minute hand

$$30H - \frac{11}{2}M$$

where,
H is the hour and M is the minute.

If answer is more than 180°, subtract the answer from 360° to get interior angle. Answer can be in minus sign. (Minus sign can be ignored)

Mirror Image

Subtract the given time from 12:00 or 11:60. ~~convni~~ 11:60 is more convenient. If in hour 12 is given, then take 12 as 00.

Q. Find the mirror image for 12:23.

$$\begin{array}{r} 11:60 \\ -00:23 \\ \hline 11:37 \end{array}$$

Consider 12 as 00. If in answer, in hour place comes 00, consider it as 12.

Water Image

Subtract the given time from 18:30 if ~~is~~ minute is less than 30. If minute is greater than 30, then subtract from 17:90.

If $M > 30 \longrightarrow$ ~~18:30~~ 17:90

If $M < 30 \longrightarrow$ 18:30

If answer is greater than 12:00, we can subtract 12:00 from the answer to get the answer in 12-hour format.

Crpt - Arithmetic

1. Each alphabet will have value from 0 to 9.
2. If $A=0$ then B cannot have value as 0. (A particular value can be assigned to at max one alphabet.)
3. Whenever we add two similar digits $(P+P)$, the result will always be a even number if no carry is there or odd if carry is there.

$$\begin{array}{r} P \\ +P \\ \hline \text{even} \end{array} \qquad \begin{array}{r} +1 \\ P \\ +P \\ \hline \text{odd} \end{array}$$

4. If $A+B=A$, then the value of B can be 0 or 9 (with carry; $9+1=10$) only.
5. You can have at max 10 unique characters. If more than 10 unique characters are there, then answer will be "Cannot be determined".
6. Sometimes we have to assume values. Assumption can be correct or wrong. Assumption will be wrong if any two alphabets have ~~tw~~ same values. Also keep a close eye on carry.

Q. $NINA = AGAIN - SING$, find the value of $S+I+N+G$?

$\Rightarrow$ We can write $NINA = AGAIN - SING$ as $NINA + SING = AGAIN$.

$$\begin{array}{r} NINA \\ +\ SING \\ \hline AGAIN \end{array}$$

Even if we add largest four digit number (i.e., 9999) with itself, we get answer starting from 1 (19998). So,

$A = 1$

$$\begin{array}{r} N\ I\ N\ 1 \\ +\ S\ I\ N\ G \\ \hline 1\ G\ 1\ I\ N \end{array}$$

$I+I$ is giving 1. But $I+I$ = even. Therefore 1 is getting carried by previous addition.

$$\begin{array}{r} 1 \\ N\ I\ N\ 1 \\ +\ S\ I\ N\ G \\ \hline 1\ G\ 1\ I\ N \end{array}$$

$I+I+1=1$

$2I+1=1$

$I=0$ or $I=5$ (10 gets 1 carried away)

Assume $I=5$

$$\begin{array}{r} 1 \\ +\ N\ 5\ N\ 1 \\ S\ 5\ N\ G \\ \hline 1\ G\ 1\ 5\ N \end{array}$$

$N+N=15$ (odd), therefore carry is coming ~~pron~~ from previous addition. Also ~~is~~ if $I=5$, then $I+I+1=11$, one will carry to next equation

$$\begin{array}{r} 1\ 1\ 1 \\ N\ 5\ N\ 1 \\ +\ S\ 5\ N\ G \\ \hline 1\ G\ 1\ 5\ N \end{array}$$

To give carry 1 to next equation, $1+G \geq 10$. Therefore G should be definitely 9. But by keeping $G=9$, N becomes 0 and $1+N+N = 1+0+0 \neq 15$. Hence, ~~ou~~ assumption that

I=5 is wrong.

Therefore I=0.

By keeping A=1 and I=0,

$$\begin{array}{r} N\,O\,N\,I \\ +\,S\,O\,N\,G \\ \hline 1\,G\,I\,O\,N \end{array}$$

N+N=O=even. So no carry. ~~als~~ Also in second from last equation, O+O=I. So there is definitely carry here. N+N=10 and N+N≠0

N=5

$$\begin{array}{r} 1\;\; \\ 5\,0\,5\,1 \\ S\,0\,5\,G \\ \hline 1\,G\,1\,0\,5 \end{array}$$

From this, we can say, G=4.

$$\begin{array}{r} 0\;1\;0\;\; \\ 5\,0\,5\,1 \\ +\,S\,0\,5\,4 \\ \hline 1\,4\,1\,0\,5 \end{array}$$

To make 5+S=14, S=9. Therefore

$$\begin{array}{r} 5\,0\,5\,1 \\ +\,9\,0\,5\,4 \\ \hline 1\,4\,1\,0\,5 \end{array}$$

We have values,

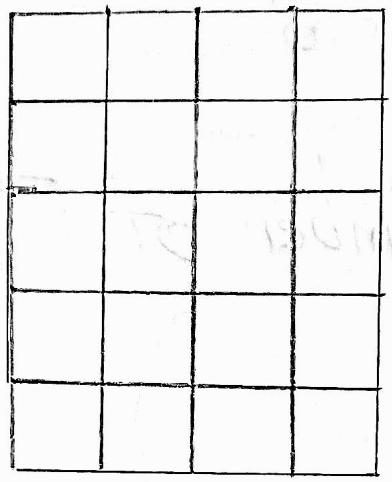
A=1, I=0, N=5, G=4, S=9

So, S+I+N+G = 9+0+5+4 = 18.

Sometimes, even by getting one or two values, we can eliminate options.

Figure Counting

Number of Squares when Row $\neq$ Column



Step 1: Count number of rows and columns

Rows = 5, Columns = 4

Step 2: Multiply ~~each row~~ number of rows with number of columns and then decrease both by 1. Repeat this step until any one is zero.

$5 \times 4 = 20$

$4 \times 3 = 12$

$3 \times 2 = 6$

$2 \times 1 = 2$

$1 \times 0 = 0$

Step 3: Add every answer. This will be final answer.

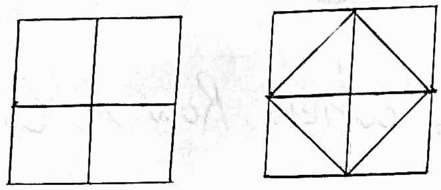
Number of squares $= 20 + 12 + 6 + 2 + 0 = 40$

Number of Squares when Row = Column

Use sum of ~~first~~ squares of first n natural numbers, where n is number of rows or columns.

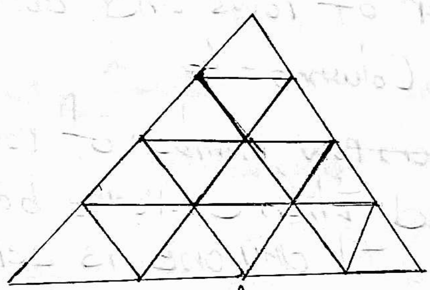
$$\text{Number of squares} = \frac{n(n+1)(2n+1)}{6}$$

Common Square Types



Number of squares = 5 6

Count Number of Triangles



Step 1: Count number of rows of triangle.
Number of rows = 4

Step 2: For the sake of calculation, create two columns, left and right. Write numbers starting from 1 till number of rows.

left	right
1	
2	
3	
4	

Step 3: On right side, in front of one, write 1. Then add it with 2, and write answer (i.e., 3) in front of 2 in write side. Add the previous answer with next ~~answer~~ number in left to find the number corresponding to that number (corresponding to number in left). Complete the tables in similar manner. Then ~~rg~~ write ~~total~~ total of right table.

left		right
1	$\longrightarrow$	1
2	+	3
3	+	6
4	+	10

Total = 20

Step 4: Add to total, numbers from right table in alternate manner while rejecting first.

left	right
1	1 ✓
2	3 ✗
3	6 ✓
4	10 ✗

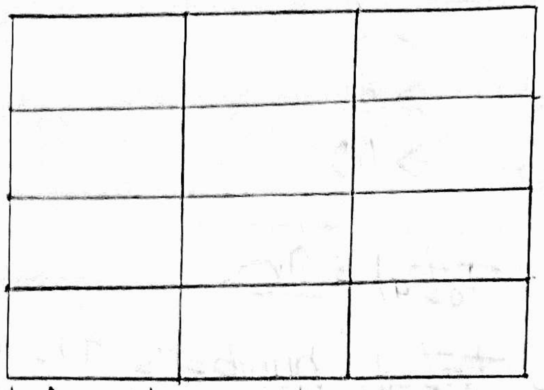
Total = 20 + 6 + 1

Step 5: Total will be the number of triangles.

Number of triangles = 27

Number of rows	Number of △s
1	1
2	5
3	13
4	27
5	48
6	78
7	118
8	170

Count Number of Rectangles (in a Grid)



Step 1: Identify the grid size

Number of rows = 4

Number of columns = 3

Step 2: Find sum of rows

Add all row numbers from 1 to 4:

$1+2+3+4 = 10$

Step 3: Find sum of columns

Add all column numbers from 1 to 3:

$1+2+3 = 6$

Step 4: Multiply both sums

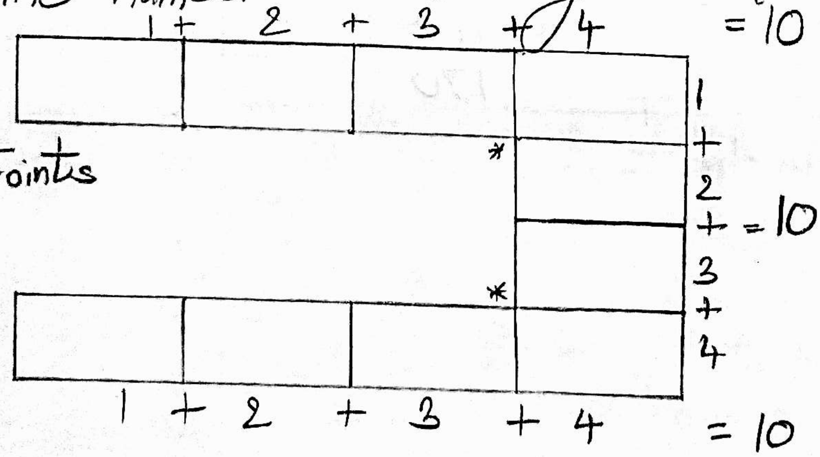
$10 \times 6 = 60$

60 is the total number of rectangles

We can also use sum of first n natural numbers:

$$\text{Sum} = \frac{n(n+1)}{2}$$

Q. Find number of rectangles:



* = Joints

Number of rectangles

= 10 + 10 + 10 − Number of joints

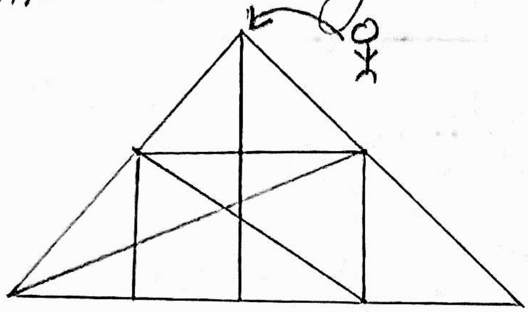
= 30 − 2

= 28

Count Number of Straight Lines

Stand on a point and then check all lines (Horizontal, Vertical, Slanting) from that point.

Q. Calculate number of straight lines.



⇒ Horizontal Lines = 2

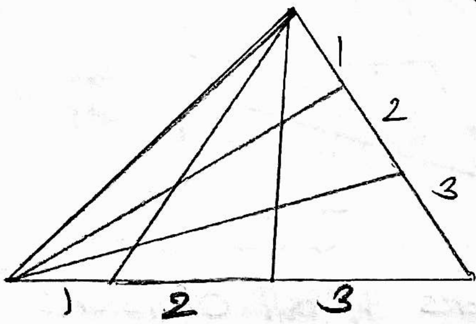
Vertical Lines = 3

Slanting Lines = 4

Total Lines = 2+3+4 = 9

Count Number of Triangles

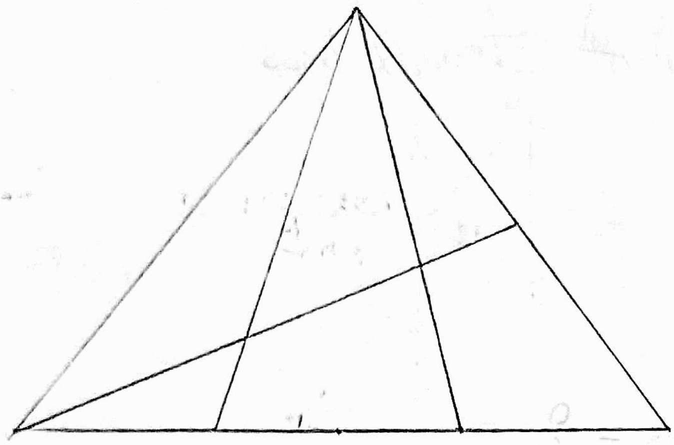
with equal divisions coming from ~~at~~ two points



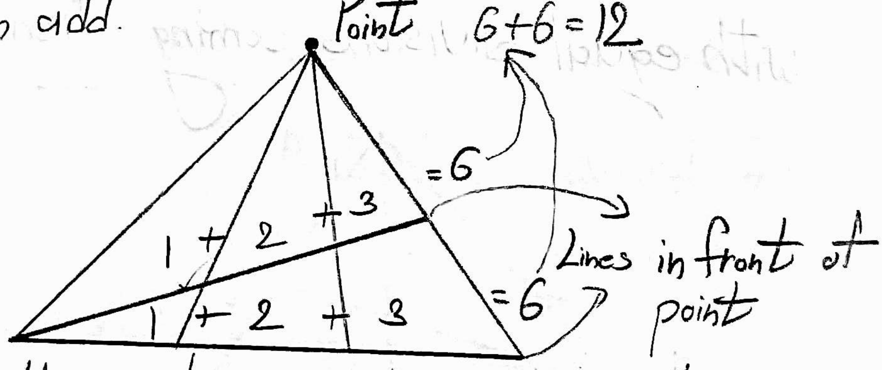
Count number of divisions on any side (any side which has been cut). The cube of the number of divisions will be the answer.

Number of triangles = $3^3 = 27$

Count number of triangles with non equal divisions coming from vertices

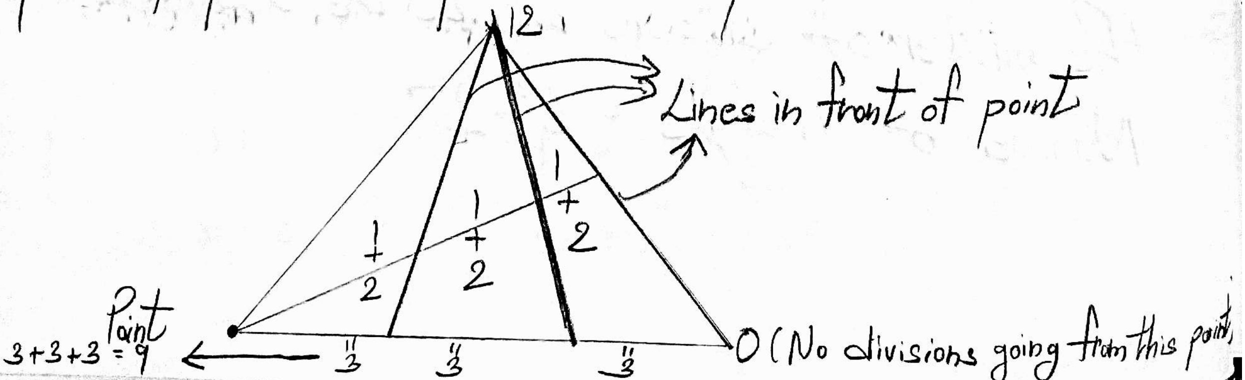


Step 1: Pick a point (vertice) and see the number of divisions from the perspective of that point. The divisions should be made by lines in front of that point. Divide the divisions into rows and give them numbers inside the row starting from 1. Then add.



Add the numbers in all divisions. Then mark that point with final sum

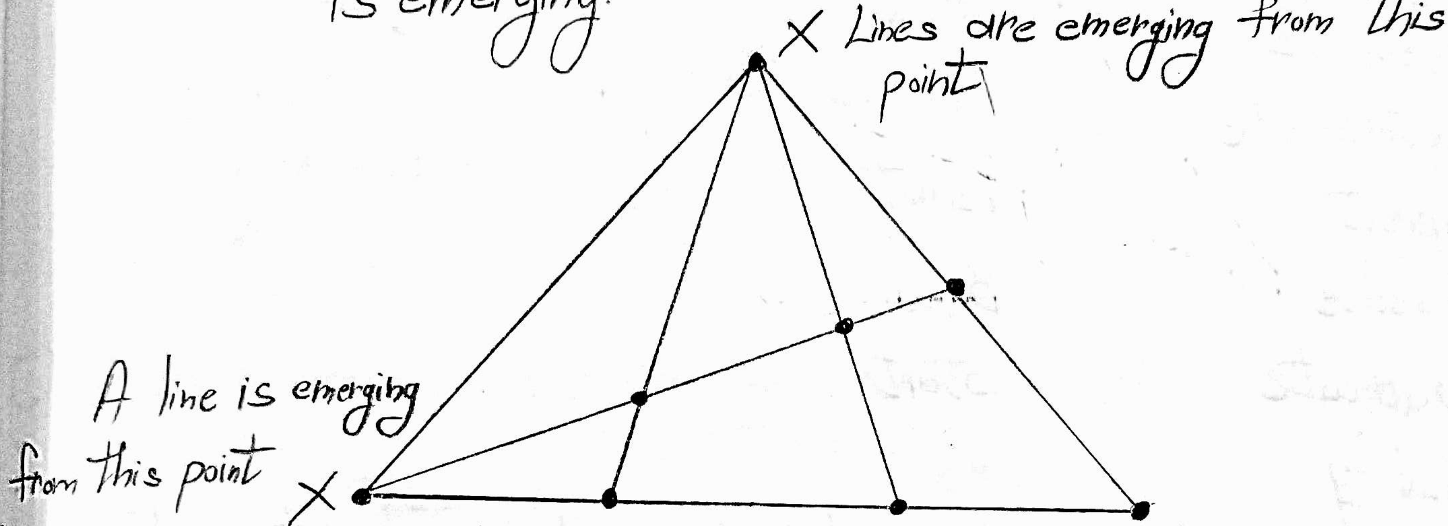
Step 2: Repeat step 1 for all points



Step 3: Add all found ~~numbe~~ marking of every vertice.

$6 \quad 12+9=21$

Step 4: Count number of points of intersection but ignore the points from where the line(s) is emerging.

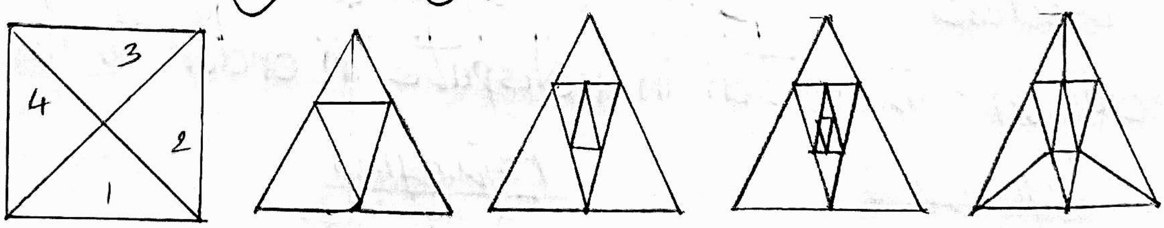


Number of ~~lines~~ points of intersection = 6

Step 5: To get the final number of triangles, subtract the sum of marking of every vertice by calculated number of points of intersection.

Number of Triangles = 21 − 6 = 15

Common Triangle Types



Number of Triangles =	8 (4×2)	5 (4+1)	9 (4+4+1)	13 (4+4+4+1)	15 (4+4+1+6)